block to measure different types of electrical quantities is as follows.

How to measure resistance?

Keep rotary switch in the position 1. Connect an unknown resistor across its input probes. The proportional current flows through the resistor, from constant current source (Battery). According to Ohm's law, a potential difference is obtained across it. This voltage is directly proportional to its resistance. That is, higher the resistance value, higher the potential difference and vice versa. This voltage is buffered and fed to A-D converter, to show in the digital display in Ohms.

How to measure AC voltage?

Keep rotary switch in position-2. Connect an unknown AC voltage across the input probes. If the voltage is above the selected range, the voltage should be attenuated and then rectified to convert it into proportional DC voltage. It is then fed to A-D converter to show in the digital display in Volts.

How to measure AC current?

Keep the switch in position-3. Current is indirectly measured by converting it into proportional voltage. Connect an unknown AC current across input probes. The current is converted into voltage proportionally with the help of I-V converter and then rectified. Now the voltage in terms of AC current is fed to A-D converter to show in the digital display in Amperes.

How to measure DC current?

Keep the switch in position-4. The DC current is also measured indirectly. Connect an unknown DC current across input probes. The current is converted into voltage proportionally with the help of I-V converter. Now the voltage in terms of DC current is fed to A-D converter to show in the digital display in Amperes.

How to measure DC voltage?

Keep the switch in position-5. Connect an unknown DC voltage across input probes. If the voltage is above the selected range, the voltage should be attenuated and then directly fed to A-D converter to show in the digital display in Volts.

The comparison between the digital and analog multimeter is given in Table 10.1.

Table 10.1 Comparison between digital and analog multimeter

S. No.	Digital Multimeter (DMM)	Analog Multimeter (AMM)
1.	Reading in digits.	Needle shows reading but the users have to find values.
2.	More accurate	Less accurate
3.	If polarity is reversed in measuring the DC actual reading with '-' sign is shown in the display.	If polarity is reversed, the needle gets reversed and subsequently the meter will become faulty.
4.	Cost is high.	Cost is lesser than DMM.
5.	If fault occurs in DMM, rectification is complicated.	If fault occurs, it may be rectified.

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